

Power and Energy Analysis of BLDC Motors: Impact of Vibration on Performance

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Research Article

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Power and Energy Analysis of BLDC Motors: Impact of Vibration on Performance

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Abstract:

This research paper focuses on the vital role of Brushless Direct Current (BLDC) motors in hybrid vehicle propulsion due to their efficiency and compact design. The primary objective is to evaluate BLDC motor performance within a vehicle chassis, with a specific focus on vibration characteristics. An innovative rotational tracker program is introduced to identify rotating machinery faults efficiently, reducing computational costs and aiding machinery selection for specific applications. The re-search involves comparing vibration characteristics of BLDC and stepper motors, using a 4kW BLDC motor to measure dis-placements along the X, Y, and Z axes within specified RPM and torque ranges while monitoring current variations during chassis installation. Findings emphasize the significance of proper motor installation in hybrid vehicle design to mitigate increased vibration levels. This study has implications for motor manufacturers and vehicle designers, underlining the im-portance of considering vibration performance during motor installation. It concludes by suggesting further research on broader RPM range BLDC motors and on-road measurements to advance understanding in the context of hybrid vehicle pro-pulsion. In summary, this research explores the connection between BLDC motors, vibration characteristics, and their impact on vehicle performance, with significant implications for the automotive industry.

Keywords: BLDC Motor, Arduino Uno, hardware connectivity via MATLAB/Simulink, power and energy performance.

1. Introduction

1.1. Background and Motivation

Vibration analysis is a well-established and indispensable tool for evaluating the condition of machinery and enabling predictive maintenance strategies. Among the various techniques employed in this field, order tracking has emerged as a valuable method for identifying faults in rotating machinery, such as motors and engines. Nevertheless, the development of software for order tracking poses significant challenges, demanding both time and computational resources.

Figure 1 presents a graphical representation of the motor's performance characteristics, highlighting the maximum power, maximum current, maximum speed, and maximum efficiency points.

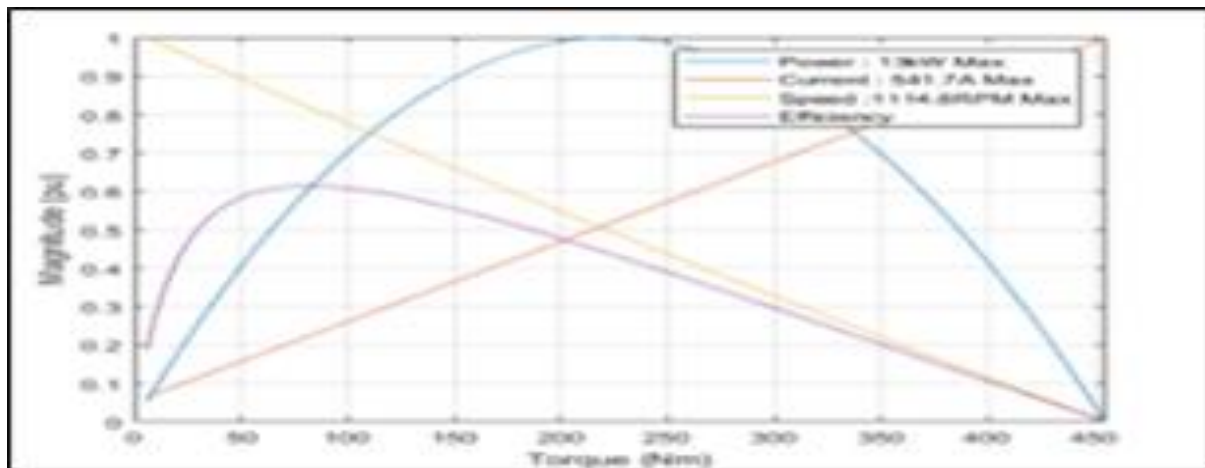


Fig. 1. Displays the performance curves of a BLDC motor using ideal input.

The most prominent feature of brushless DC motors is their maintenance-free operation. They exhibit several distinctions from traditional DC motors, one of which is their compact size. As illustrated in Figure 4, a high-performance brushless DC motor accommodates high-performance magnets within a smaller space compared to conventional DC motor rotors. To optimize space efficiency, the rotor lacks magnets, thereby allowing for downsizing in the corresponding space. This higher magnetic flux density results in enhanced motor efficiency while minimizing heat generation.

Among all motor components, the coil generates the highest amount of heat. In the case of a DC motor, the coil is located inside the motor housing, enveloped by the surrounding air, which hinders the dissipation of heat from the coil. Conversely, in a brushless motor, the coil is positioned on the stator, thereby improving the motor's heat dissipation performance. The decisive advantage of brushless motors lies in their significant design flexibility. For example, you can design a flat rotor, position the rotor on the outer periphery of the motor, or opt for an elongated shape, depending on the intended application of the motor.

The literature review provides a comprehensive overview of the research in the field of BLDC motors, vibration analysis, and control strategies, offering valuable insights into the optimization of motor performance and efficiency.

1.2 Mathematical Model

The mathematical model presented in this paper illustrates the electrical parameters of the motor in matrix form to evaluate the voltage states a and b of the motor. Manufacturers of BLDC motors specify a parameter known as the back EMF constant, which can be utilized to estimate the back EMF for a given speed. The potential across a winding can be calculated by subtracting the back EMF value from

the supply voltage. The back EMF is computed based on the difference between the supplied voltage and the current loss through the resistance. It appears that the back EMF constant is approximately 2.065 V (ln-RMS)/rpm. The inductance (L) is calculated according to equation 1, where we have used the previously determined values of resistance (R) and frequency (f).

$$L = \sqrt{2 * \pi * f} \quad (1)$$

$$\begin{bmatrix} V_a \\ V_b \end{bmatrix} = \begin{bmatrix} R_a & 0 \\ 0 & R_b \end{bmatrix} \begin{bmatrix} i_a \\ i_b \end{bmatrix} + \begin{bmatrix} L_a & 0 \\ L_b & 0 \end{bmatrix} \begin{bmatrix} \frac{di_a}{dt} \\ \frac{di_b}{dt} \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \end{bmatrix} \quad (2)$$

The torque is the sum of the cogging torque and electromotive force (EMF), expressed as Equation 2. Inductances and EMF can be determined through finite element (FE) analysis or obtained from the catalog. Here, T_{cog} represents the cogging torque.

$$T = \frac{1}{\omega} (e_a i_a + e_b i_b) + T_{cog} \quad (3)$$

$$T_E(t) - T_L(t) = J \frac{d\omega(t)}{dt} + B * \omega(t) \quad (4)$$

Where:

T represents the total torque

T_{cog} represents the cogging torque.

T_L represents the load torque in N-m.

J stands for rotor inertia in [kgm²]

B represents the damping constant

2. Literature Review

The literature on Brushless Direct Current (BLDC) motors and their interconnection with vibration analysis and control techniques provides valuable insights into the field:

Vibration Analysis and Control Techniques: The vibration analysis of BLDC motors has been conducted using various methods. Notably, the use of the 3-Dimensional Finite Element Method (3-D FEM) to analyze the impact of Z-axis eccentric force due to commutated current on BLDC motor vibrations has been explored [1]. Additionally, state-of-the-art methods such as Terminal Voltage Sensing, Third Harmonic Voltage Integration, Terminal Current Sensing, Back-EMF Integration, and PWM strategies have been applied for analysis, often in conjunction with Artificial Neural Networks [2]. Techniques such as the Proportional-Integral (PI) controller and wavelet-based denoising have been employed to achieve precise torque tracking and reduce vibration [3].

Speed and Torque Control: BLDC motor drives for speed and torque control have been a significant focus of research. The use of semiconductor switching devices like MOSFETs and JFETs has been investigated to ensure satisfactory performance in speed control [4]. Various control strategies, including ANFIS (Adaptive Neuro-Fuzzy Inference System), PI, and PID controllers, have been studied to minimize ripples in torque and speed while maintaining admissible current control [5]. The PI controller has shown improved performance when the load is applied gradually [6].

Optimization and Performance Enhancement: Addressing stator magnetic saturation and optimizing motor cogging torque has been a key research area to enhance BLDC motor performance [7]. Predicting torque and speed values based on back-EMF waveforms has been simulated to verify BLDC motor performance [8]. Simulation studies for BLDC motor speed control, comparing intelligent controllers (ANFIS) with conventional controllers (PI, PID), have been conducted for performance analysis [9]. The modeling of BLDC motor drives with control systems has been evaluated for speed and current to ensure satisfactory performance [10].

Experimental Analysis of Vibration and Noise: Experimental analysis of vibration and acoustic noise in BLDC motors has been carried out using digital pulse width modulation (PWM) based control

techniques [11]. Electronically commutated switches for rotor position detection have been used for motor speed control [12]. Fault isolation, based on fault signature tables, has been tested under noisy motor conditions [13]. Detailed analysis using Ansys software has been employed to identify the main sources of vibration in BLDC motors, including the participation of individual parts like stators and rotors [15].

Motor Efficiency and Propeller Efficiency: The review highlights the interplay between motor efficiency and propeller efficiency, emphasizing that motor efficiency is influenced by operating conditions, particularly speed and torque levels. It discusses the trade-offs between motor efficiency and the use of larger, more efficient propellers to avoid the need for a gearbox, which can introduce complexity and additional heat loss.

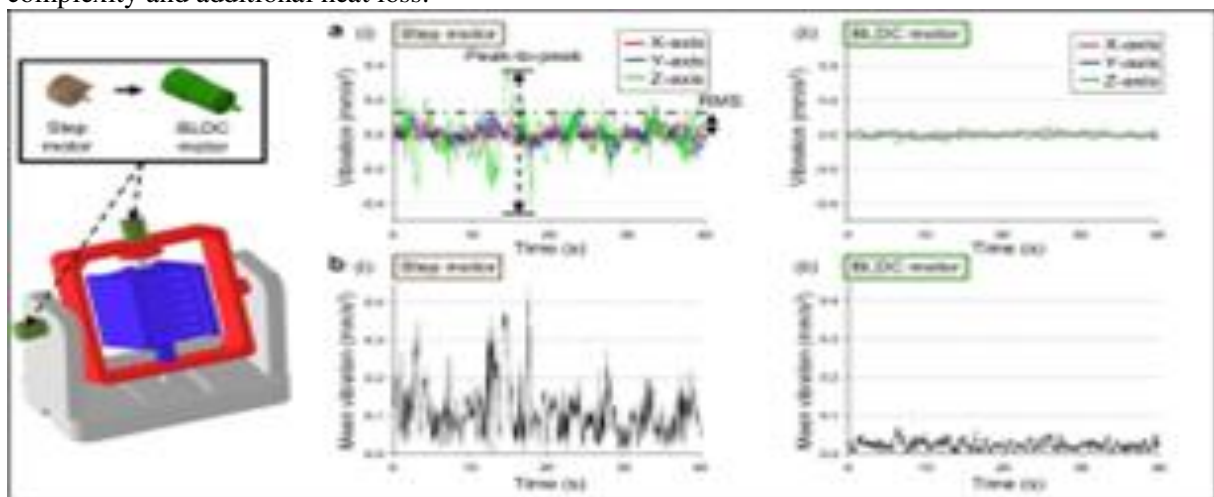


Fig. 2. Comparison of Vibration Waveforms for Stepper and BLDC Motors Over Time.

Maximum Power and Efficiency Analysis: The literature introduces key performance parameters, such as maximum power, maximum torque, and motor efficiency. It emphasizes the importance of identifying the motor's peak efficiency point, where it can produce the most power and torque with the least amount of wasted energy. Among all motor components, the coil generates the highest amount of heat. In the case of a DC motor, the coil is located inside the motor housing, enveloped by the surrounding air, which hinders the dissipation of heat from the coil. Conversely, in a brushless motor, the coil is positioned on the stator, thereby improving the motor's heat dissipation performance. The decisive advantage of brushless motors lies in their significant design flexibility. For example, you can design a flat rotor, position the rotor on the outer periphery of the motor, or opt for an elongated shape, depending on the intended application of the motor.

2.1 Limitations

The study investigates limitations related to the design and operation of brushless DC motors. One limitation pertains to torque non-uniformity, which results from the sequential switching of coil currents to achieve smooth rotation. Increasing the number of coils to match traditional DC motors requires costly semiconductor devices for individual coil current control. Most brushless DC motors use three coil groups to reduce costs, leading to torque variations. Additionally, cogging, caused by magnetic attraction between rotor magnets and stator teeth, affects motor performance and necessitates a thorough magnetic circuit analysis. The study also addresses motor efficiency, with a focus on operating below the maximum power curve for extended flight time and endurance. These limitations have implications for motor design and performance optimization in various applications.

2.2 Research Investigations

In investigation, it has been observed that shorter wave-length waves are generated due to cogging torque. Cogging torque in motors is undesirable, particularly at low speeds, as it results in undesirable fluctuations in the motor's output torque. However, at higher speeds, the rotor's inertia contributes to the smoothing of these fluctuations, rendering them less noticeable.

This paper has delved deeper into the implications of these wave characteristics, particularly as they relate to machinery performance and cogging torque in motors. Our findings provide valuable insights into the design and selection of motors to optimize machinery performance and minimize undesirable vibrations.

The following table summarizes the key differences between the two motors:

Table 1. Key differences between the two motors.

Feature	DC Motor	Brushless DC Motor
Rotor	The coil rotates around a permanent magnet	A permanent magnet rotates around the coil
Commutation	Mechanical, using brushes	Electronic, using a controller
Heat dissipation	Air space restricts heat propagation to the surface	High-performance magnet provides a good heat dissipation effect
Advantages	Simple, rugged, low cost	Efficient, quiet, long life
Disadvantages	Requires brushes, which can wear out	More complex, and requires a controller

The DC motor is a simpler and more rugged motor, but it is also less efficient and has a shorter lifespan. The brushless DC motor is more efficient and has a longer lifespan, but it is also more complex and requires a controller.

Figure 3. illustrates the different parts of a DC motor and a brushless DC motor. The DC motor has a coil that rotates around a permanent magnet. The brushless DC motor has a permanent magnet that rotates around a coil. The coil in both motors generates heat, but the air space around the coil in the DC motor restricts the heat propagation to the surface. The brushless DC motor has a high-performance magnet that provides a good heat dissipation effect. The figure also shows the different ways that the coils are commuted in the two motors. In the DC motor, the brushes contact the commutator to switch the current in the coil. In the brushless DC motor, the commutation is done electronically by a controller

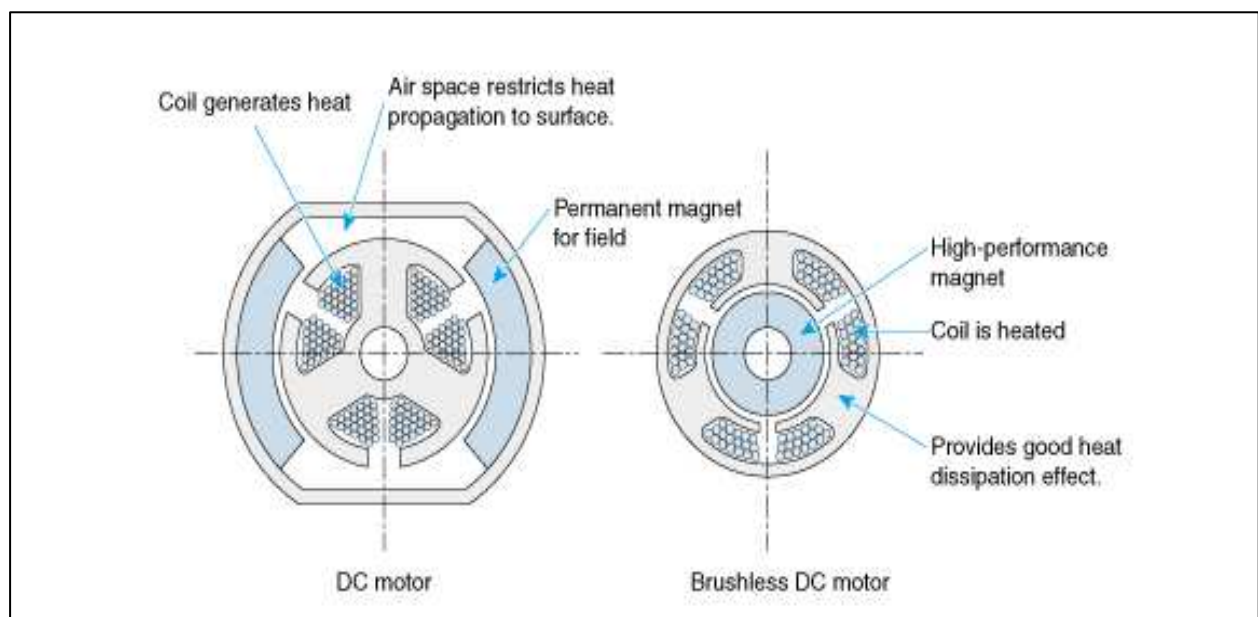


Figure 3: Comparison of Cross-section between a General DC Motor and a Brushless DC Motor.

2.3 Problem Statement

In Figure 1, represents a comparison of vibration characteristics between two common types of motors, the Brushless DC (BLDC) motor and the stepper motor. The graph illustrates that the BLDC motor exhibits higher vibration amplitudes, peak-to-peak vibration, and Root Mean Square (RMS) vibration compared to the stepper motor. This difference can be attributed to the BLDC motor's higher rotational speed. Such findings have practical implications for selecting the most suitable motor for specific applications, as motor vibration can significantly affect the performance and reliability of machinery.

2.4 Research Objectives

Furthermore, the relationship between wavelength, frequency, and energy is a fundamental concept in the context of wave phenomena. As wavelength increases, frequency and energy decrease, and conversely, as frequency decreases, wavelength lengthens. These principles apply to two broad categories of waves: mechanical and electromagnetic. Notably, shorter wavelengths, characterized by higher frequencies, carry increased energy and exhibit a higher likelihood of interactions with the surrounding medium. Consequently, waves with shorter wavelengths entail greater energy consumption.

3. Methodology

3.1 BLDC Motor Selection and Characteristics

This study employed an experimental approach to investigate Brushless Direct Current (BLDC) motor characteristics. The BLDC motor selected for experimentation was analyzed for key parameters including back electromotive force (EMF) constant (K_E) and torque constant (K_T). These characteristics play a crucial role in motor performance and efficiency.

3.2 Data Acquisition Setup

A comprehensive data acquisition setup was implemented to capture relevant motor data. The setup included a Hybrid Electric Vehicle (HEV) equipped with a data acquisition tool capable of recording fuel and electricity usage over specific time intervals. Additionally, vehicle position data was logged during driving to analyze the influence of driving conditions on motor performance.

3.3 Vibration Measurement Techniques

Vibration analysis was conducted to assess the mechanical aspects of the BLDC motor. A piezo accelerometer/vibration sensor was integrated into the setup to capture vibration data. The acquired vibration data was essential for understanding motor behavior and identifying potential issues such as cogging.

3.4 Energy and Power Analysis Methods

Energy and power analysis methods were applied to evaluate the performance of the BLDC motor. Simulink models were utilized to simulate the motor's energy and power output, allowing for a comprehensive assessment of its efficiency and operating characteristics.

3.5 Experimental Design

The experimental design encompassed the daily operation of HEVs under typical driving conditions in the Jakarta, Bogor, Depok, Tangerang, and Bekasi areas. These regions were chosen due to their congested road conditions and the presence of toll roads, reflecting real-world driving scenarios. The consistently hot tropical climate in the region necessitated the continuous use of air conditioning, which was factorized into the study.

The data collected during these experiments were analyzed to evaluate the potential of hybrid vehicles to reduce fuel consumption, thus contributing to lower air pollution levels. Furthermore, the analysis included the investigation of torque variations, commonly referred to as 'cogging,' and how they relate to the motor's rotational position.

4. Experimental Investigation

4.1 Data Acquisition

The CPU controls motor speed by adjusting the PSU voltage (ranging from 6 to 16 V) and managing the ESC through a throttle PPM signal. The DAQ measures two signals: a vibration signal from the sensor and a voltage signal from a motor phase (used to calculate the motor's rotational frequency).

The physical setup was far from ideal, but it was adequate for experimentation. In a more 'real' setup, greater care would need to be taken with cabling and connections to prevent noise leakage. Due to the rapid motor acceleration, I had to secure it to a more rigid structure to prevent it from spinning around on the bench. The small size of the motor made it challenging to attach the sensor directly to it, so I ended up attaching it to the structure to which the motor was fastened (though this was less than ideal, as the plastic cable ties dampened the motor vibrations).

The DAQ (NI myDAQ) supports a maximum sampling frequency of 200 ksp/s at 16 bits within the range of ± 10 V. Therefore, in this setup of experiment voltage divider is used for the motor phase signal analysis. For the ESC with the BLHeli firmware, disabled its PID control loop and set the minimum and maximum PPM throttle values to 1000 and 2000 ms, respectively, to allow for finer speed control. To confirm that all connections were correctly configured and that the motor was securely fastened and conducted a linear acceleration and deceleration test.

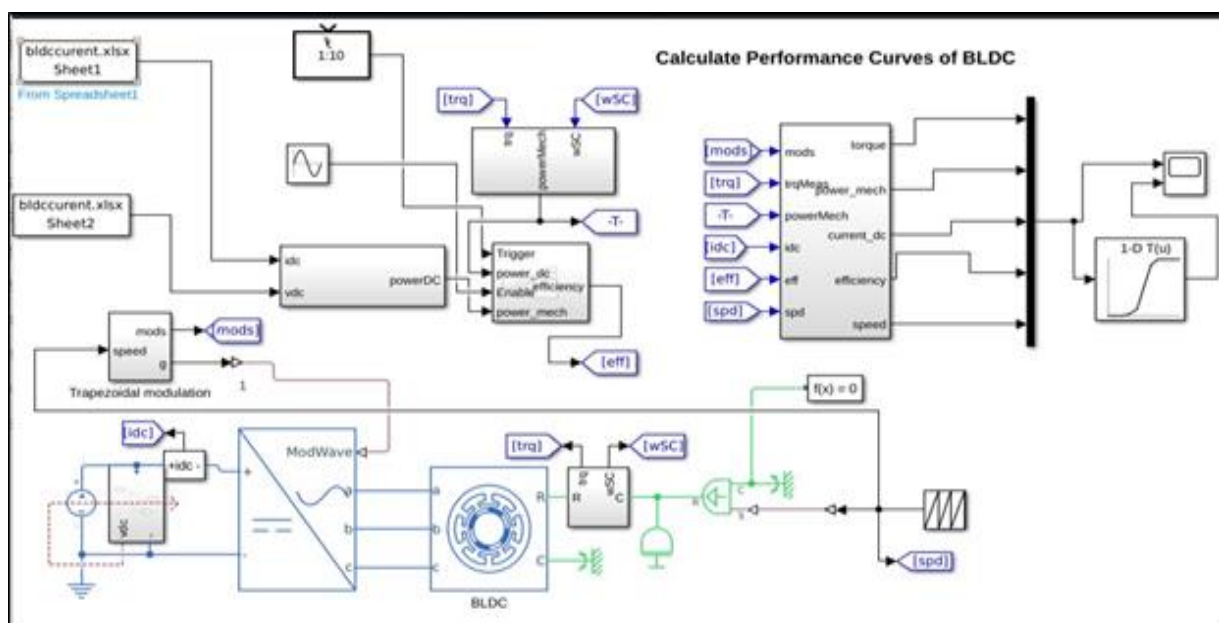


Fig. 4. MATLAB/Simulink model of a Brushless DC (BLDC) motor, illustrating its energy and power output in the Scope.

4.2 Motor Control

In Hybrid Electric Vehicles (HEVs), Brushless DC (BLDC) motors play a crucial role, operating with a rotor featuring permanent magnets and a stator housing winding, controlled by electronic commutation. Typically, three Hall sensors detect rotor position for precise commutation. The effectiveness of BLDC motors depends on the correct application of voltage waveforms between motor phases, ensuring that the stator and rotor flux angles remain close to 90° for optimal torque

generation. Commutation relies on a relatively simple position sensor, often three Hall effect sensors positioned in alignment with the back electromotive force (EMF). Sensorless control uses back EMF zero-crossing detection for commutation. This study explores the efficiency calculations for BLDC motors, offering practical insights into their implementation through experimentation

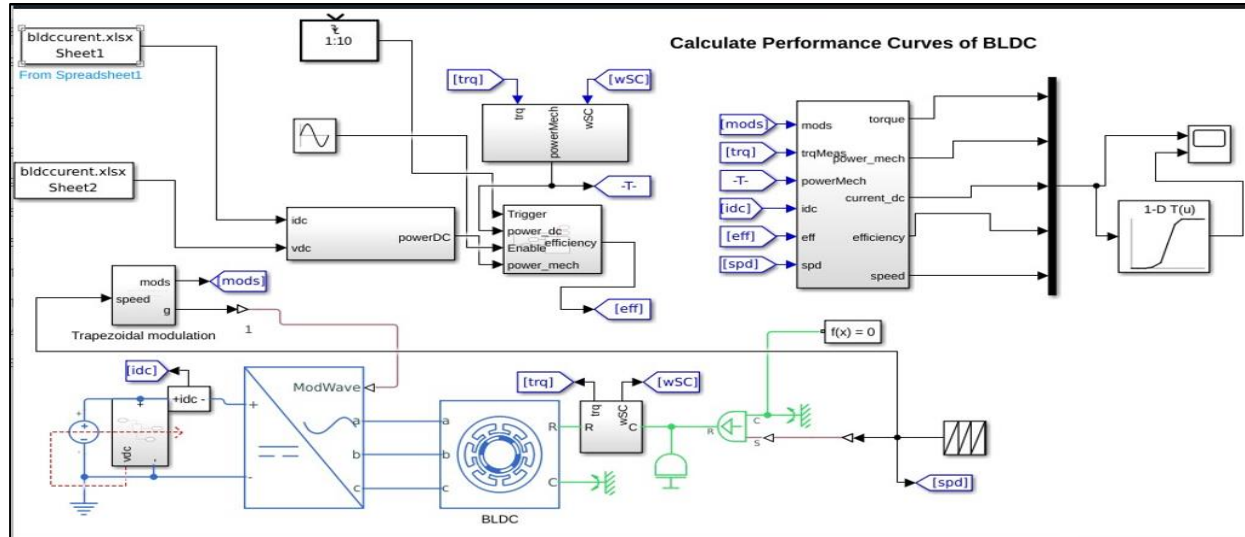


Figure 4 displays a Simulink model of a Brushless DC (BLDC) motor, illustrating its energy and power output in the Scope.

2.1 Parameters for BLDCM Experimentation

The modeling of a Brushless Direct Current (BLDC) motor's motor drive model can be developed in a manner like that of a three-phase synchronous machine. Since its rotor is equipped with a permanent magnet, certain dynamic characteristics differ. The flux linkage from the rotor depends on the magnet. Like any typical three-phase motor, one of the structures of the BLDC motor is powered by a three-phase voltage source. Here, K_E represents the back electromotive force (emf) constant, and ω represents the mechanical speed of the rotor. In the equation above, K_E is represented with a capital letter 'E' since we used a lowercase letter 'e' in the previous equation. The permanent magnet also influences the produced torques due to the trapezoidal flux linkage, where K_T represents the torque constant. The resulting torque, T_E , can be calculated using the following expressions, in accordance with Newton's second law of motion for the rotor.

2.2 Experimental Setup

The BLDC motor driving circuit is simulated as shown in Fig. 2, where V represents the input voltage, I is the input current, e is the back-EMF, R is the resistance of the armature, L is the inductance, and ω is the rotating speed as shown in Eq. (1) of the driving circuit of the 2-phase BLDC motor.

The hardware unit Arduino is connected to the Simulink model through a USB RS232 cable with Arduino driver installation (CH240). The specimen BLDC motor sample is connected to the Arduino

unit as depicted in Fig. 5. The Simulink model of an Arduino Mega program for a PID controller. PID stands for Proportional-Integral-Derivative, and it is a control loop feedback mechanism that is used to regulate the output of a system. In this case, the system is a BLDC motor (brushless direct current motor). The model can be used to simulate the performance of the PID controller and to tune the parameters of the controller.

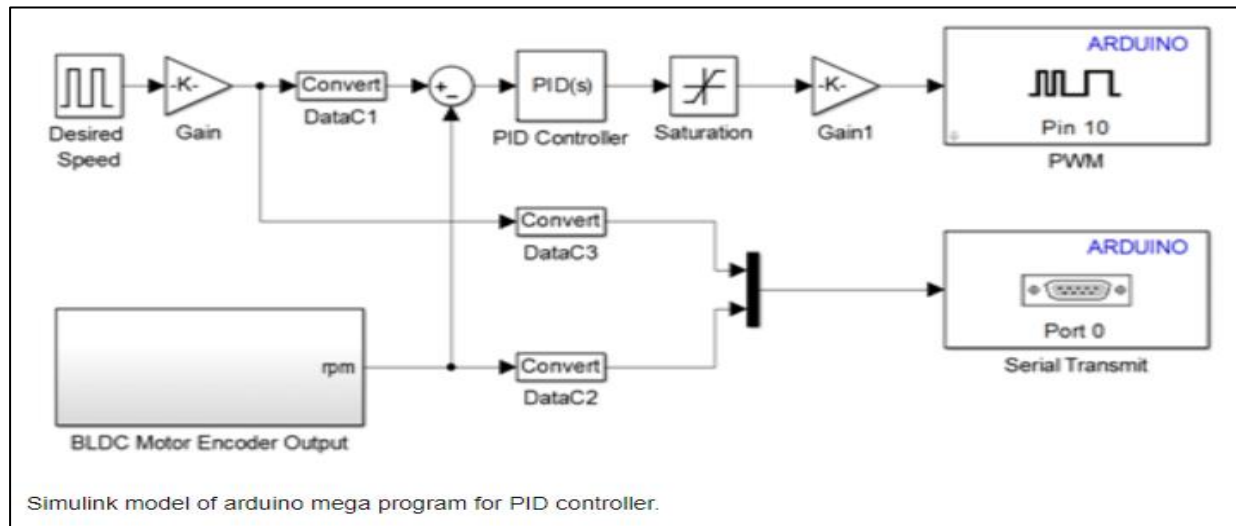


Figure 5: BLDC Motor Connectivity to Simulink Model with Arduino Unit Data Acquisition.

4.3 Signal Analysis

Signal Analysis is pivotal in electrical engineering. A line-ar sweep lasting approximately 180 seconds, ranging from 20 Hz to 200 Hz, distinguishes between asynchronous and synchronous vibration components. This sweep reveals spectrum components influenced by the shaft's rotational frequency and those unaffected, aiding in the identification of damping at specific frequencies. Rotational tracking

data was collected to create a rotational frequency curve.

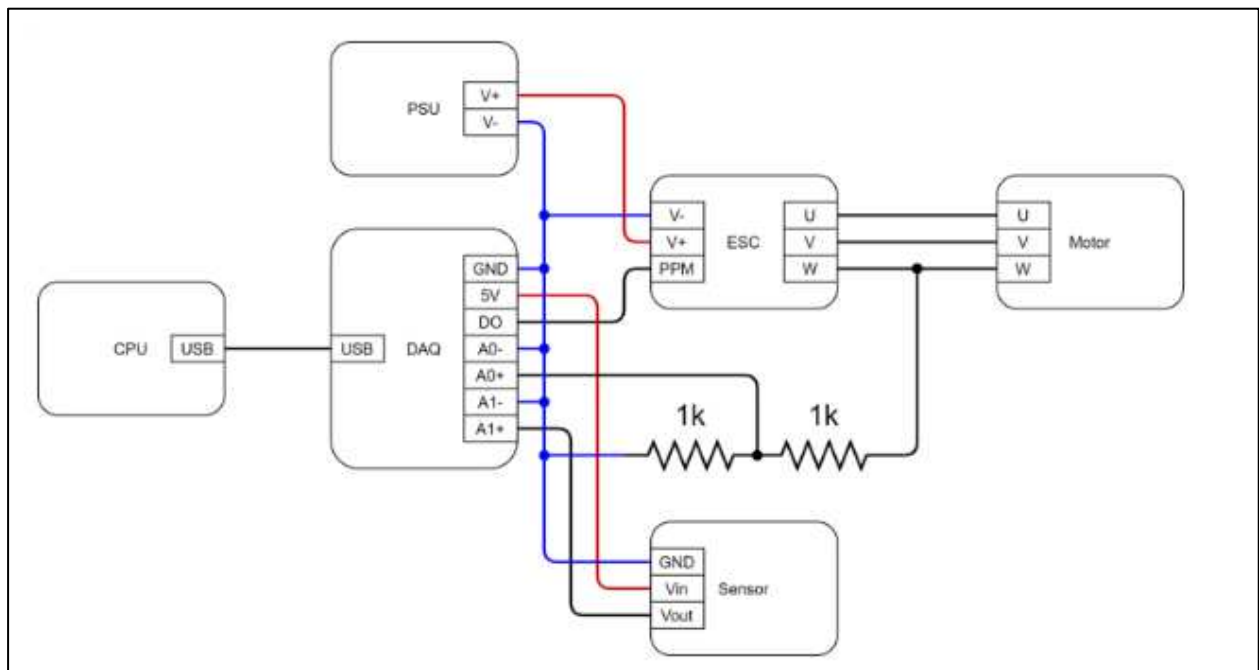


Fig. 6. Connection of a BLDC Motor to a Laptop Using an Arduino Unit (Data Acquisition - DAQ)

4.4. Experimental Setup

The experimental setup for BLDC Motor vibration velocity measurement, energy, and power performance evaluation is illustrated in Figure. The setup includes a solar panel powering the BLDC Motor through a photocell, controlled by an Electronic Speed Control Unit, connected to a MATLAB/Simulink model via an Arduino Unit.

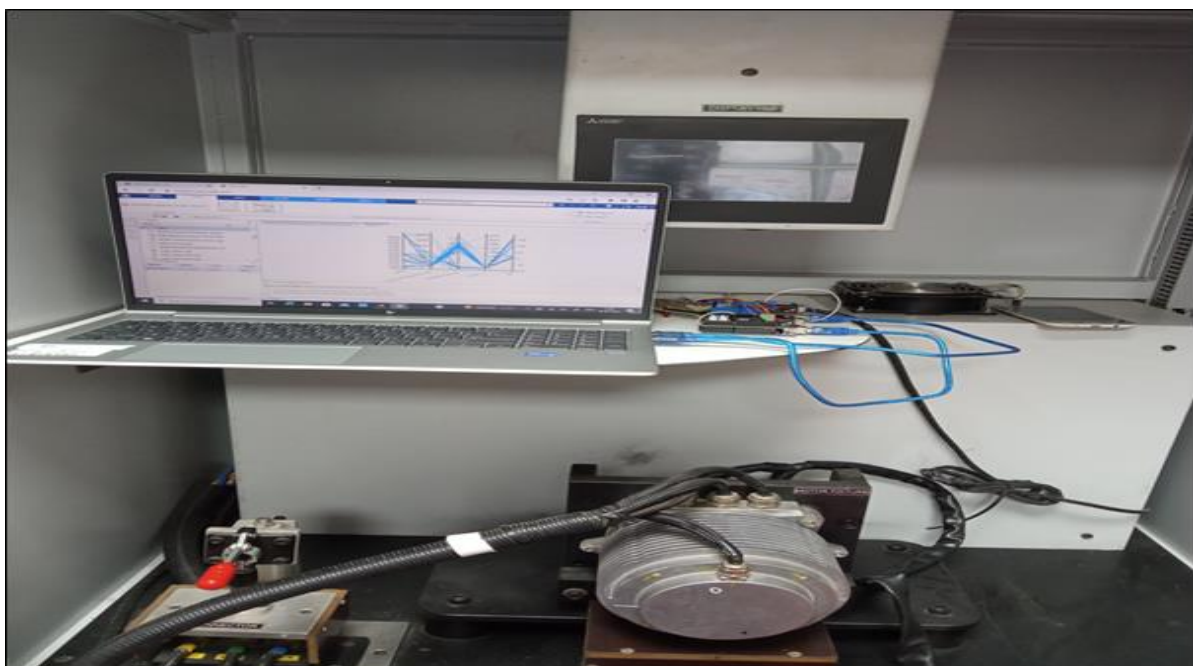


Fig. 7. Experimental Setup for BLDC Motor Vibration Testing

4.5. Graphical Presentation of Findings

Figure 8. shows the graphical results of the BLDCM (Brushless DC Motor) Energy and Power Performance curve from 1280 to 3800 RPM, and then from 3800 to 7500 RPM. It illustrates the trajectory path of current and power input to Energy, power, torque, and efficiency at various speed ranges.

The graph can also be used to select the right BLDC motor for a particular application. For example, if we need a motor that can produce a lot of torque at low RPMs, we should choose a motor with a high maximum torque line. If we need a motor that can run at high RPMs, we should choose a motor with a high maximum RPM line.

Figure 8 displays the graphical results of BLDC Motor Energy and Power Performance from 1280 to 3800 RPM and from 3800 to 7500 RPM. It charts current, power input, energy, power, torque, and efficiency at various speed ranges. This graph aids in selecting the appropriate BLDC motor for specific applications based on torque and RPM requirements.

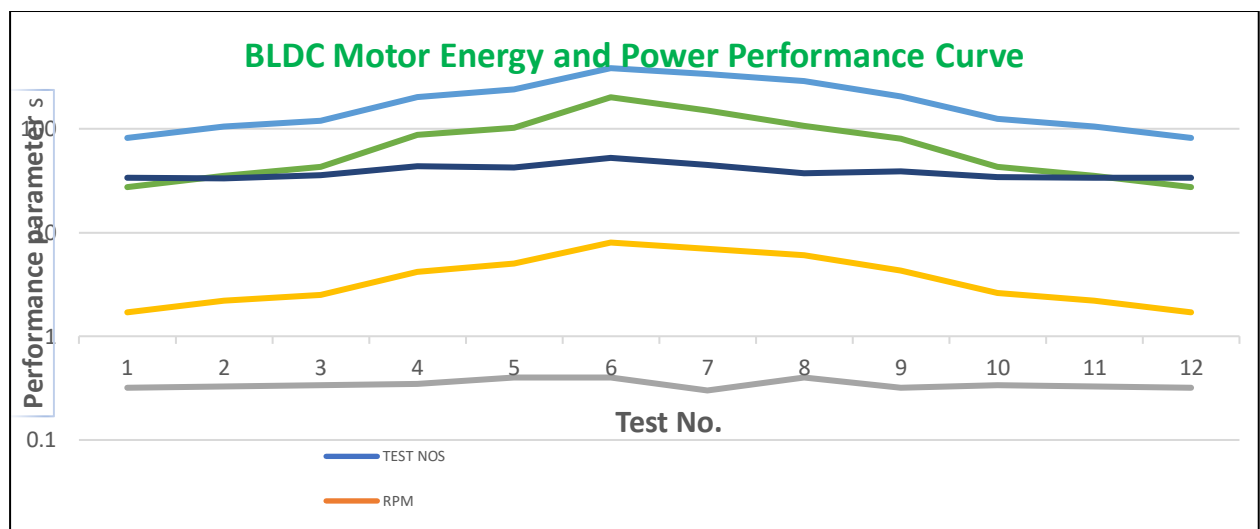


Figure. 8 The performance of a Brushless DC Motor (BLDCM) in terms of energy and power across two distinct RPM ranges, spanning from 1280 to 3800 RPM and from 3800 to 7500 RPM.

BLDCM's energy and power performance is from 1280 to 3800 RPM. This plot showcases the relationship between current, power input, and parameters like energy, power, torque, and efficiency at various speeds. The point of inter-section of the torque, current, and power lines represents the motor's peak efficiency, with efficiency decreasing as it deviates from this point. While the specific values are

motor-specific, the underlying principles apply universally to BLDC motors.

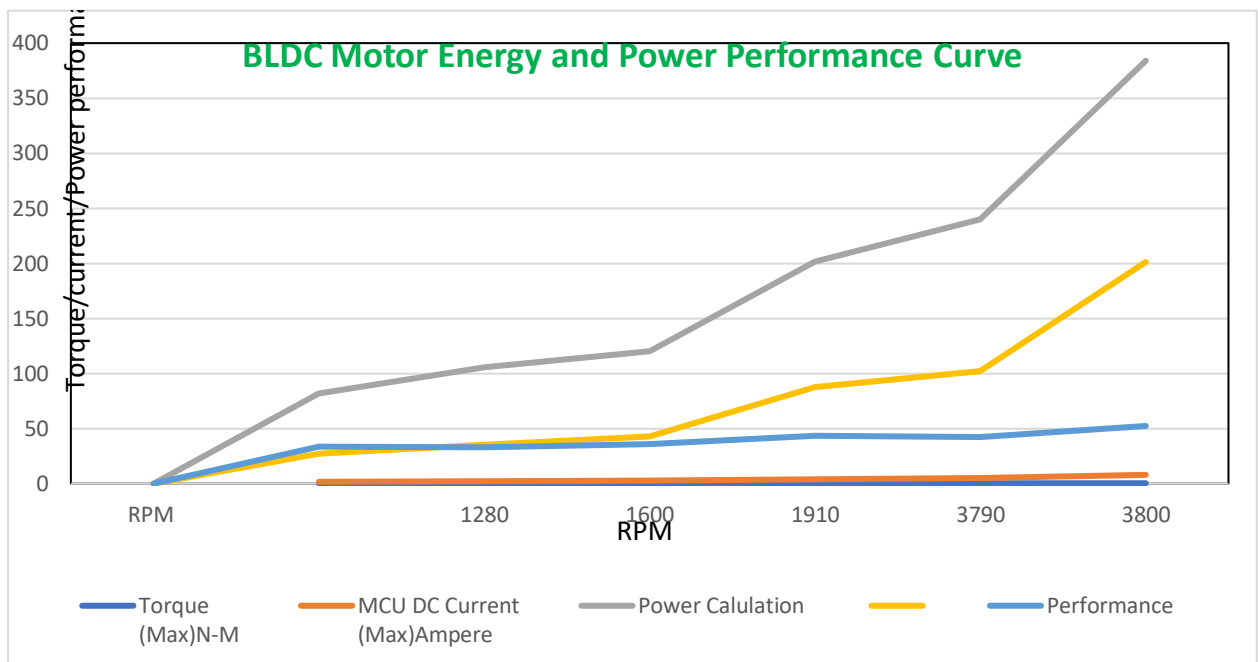


Fig. 9. The performance of a BLDCM (Brushless DC Motor) in terms of energy and power across two distinct RPM ranges: from 1280 to 3800 RPM and from 3500 to 7500 RPM

Figure. 10 offers a 3D representation of Mount Bruno's elevation, illustrating the vibration velocity of a BLDCM in millimeters per second (MM/sec) within the 1280 to 3800 RPM range in the X, Y, and Z directions

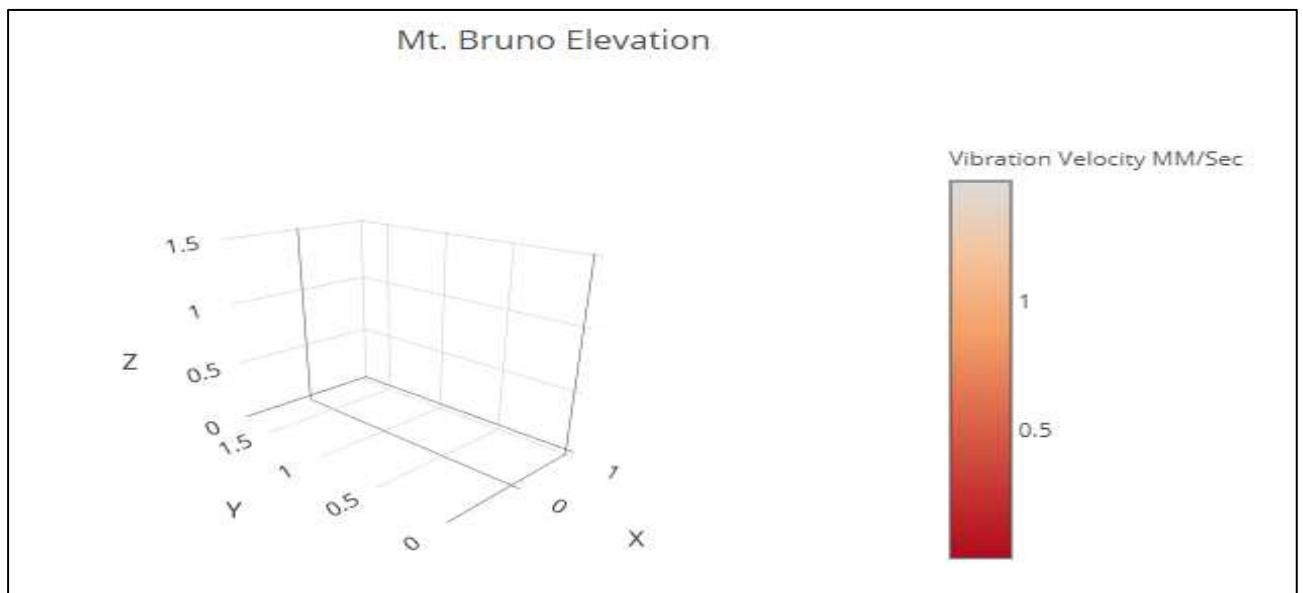


Fig.10. 3D Plot of Bruno Elevation for BLDC Motor Vibration Data

In Figure. 11, we observe velocity measurements of BLDCM vibrations in MM/sec at two RPM ranges, namely 1280 to 3800 RPM and 3800 to 7500 RPM in the X, Y, and Z directions. Notably, vibration

amplitude is significantly elevated between 4000 and 6000 RPM

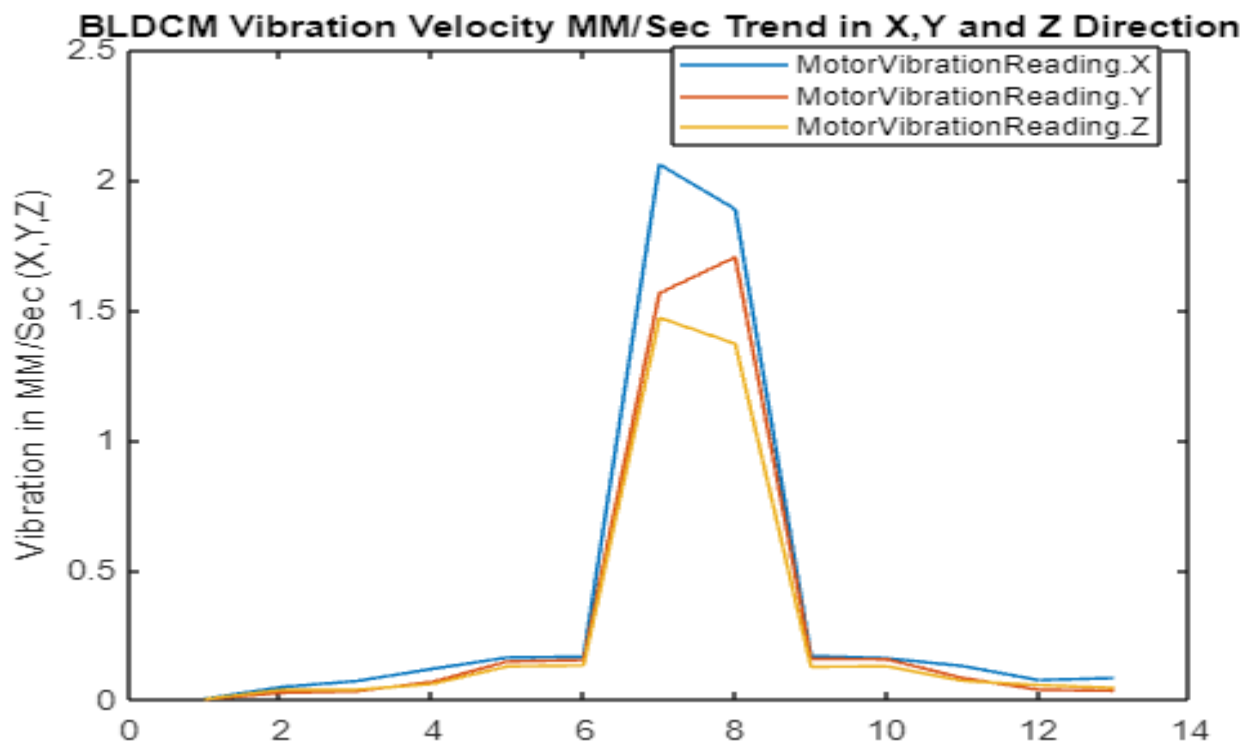


Figure 11 displays the trend of vibration velocity (in mm/sec) for a BLDCM in the X, Y, and Z directions.

Figure. 12 showcases MATLAB-generated graphical plots illustrating the Energy and Power Performance curve of a BLDCM from 1280 to 3800 RPM. This simulation portrays the impact of current and power input on energy, power, torque, mechanical power output, and overall efficiency across speed ranges. The curve demonstrates the RPM-torque relationship and the effect on electrical and mechanical power, as well as efficiency. Notably, efficiency is highest at low RPMs and torques, gradually declining as RPM and torque increase, primarily due to increased losses at higher RPMs. This curve aids in motor selection and performance optimization for specific applications, with logarithmic scales on the axes. It's essential to note that the curve's characteristics depend on motor design, load, and ambient temperature.

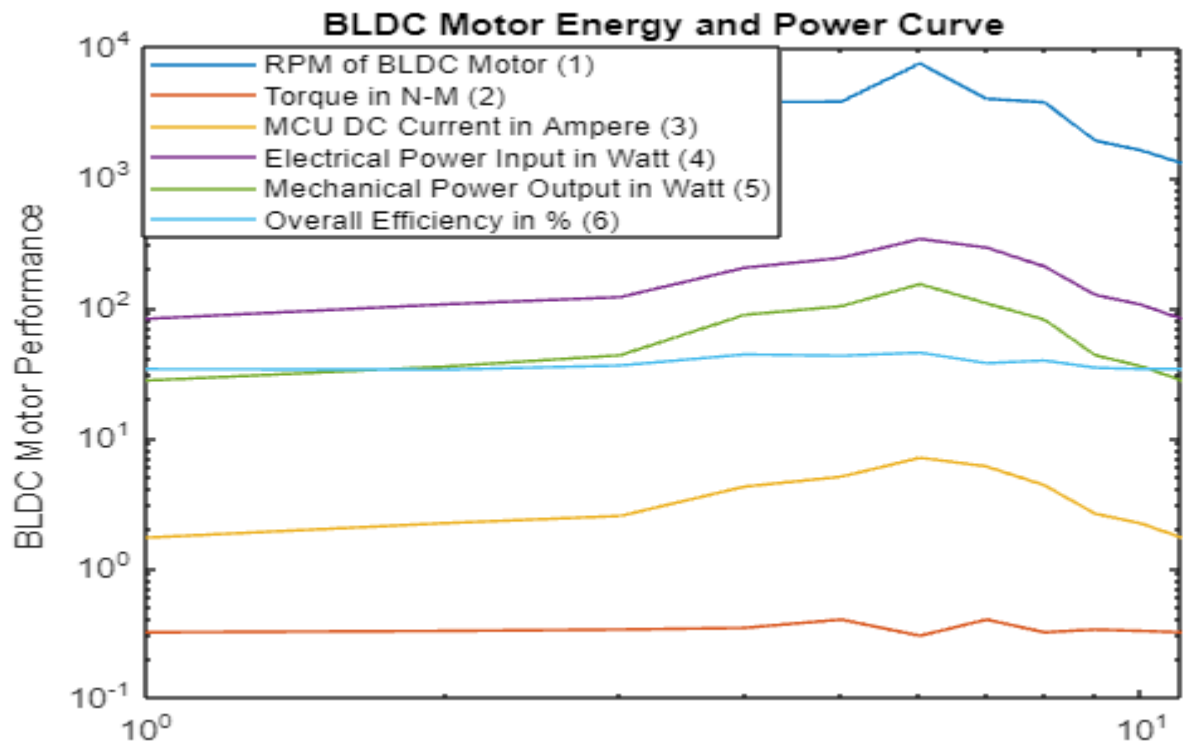


Figure 12. Energy and Power Performance Curve of a BLDCM

In Figure. 11, we observe velocity measurements of BLDCM vibrations in MM/sec at two RPM ranges, namely 1280 to 3800 RPM and 3800 to 7500 RPM in the X, Y, and Z directions. Notably, vibration amplitude is significantly elevated between 4000 and 6000 RPM in Figure. 11, we observe velocity measurements of BLDCM vibrations in MM/sec at two RPM ranges, namely 1280 to 3800 RPM and 3800 to 7500 RPM in the X, Y, and Z directions. Notably, vibration amplitude is significantly elevated between 4000 and 6000 RPM.

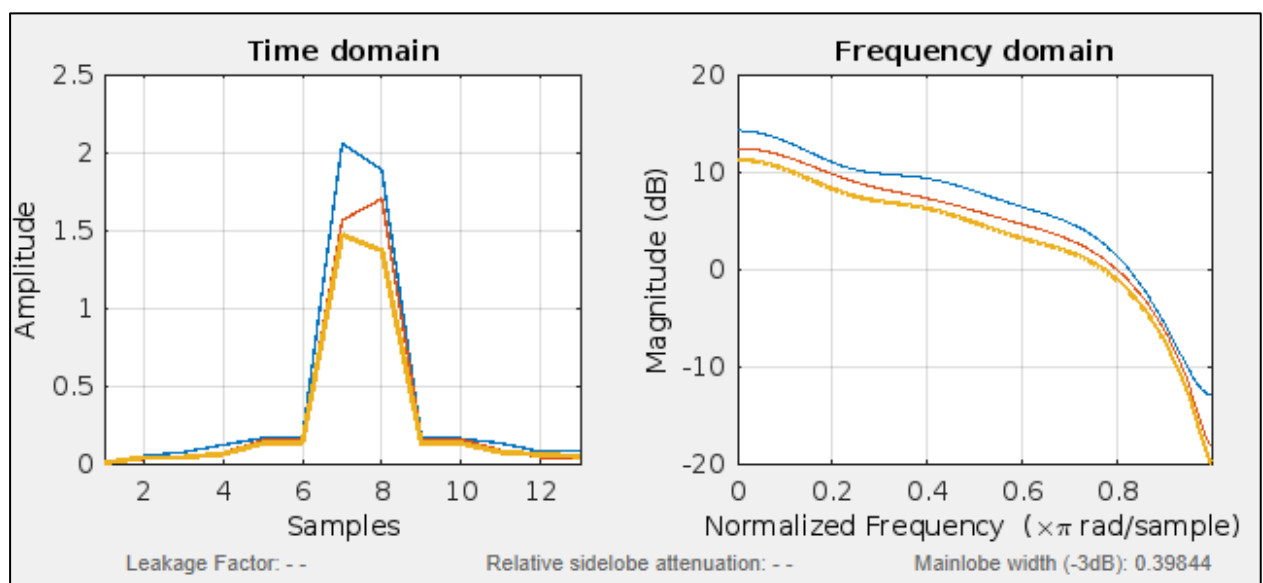


Figure 13 illustrates the vibration amplitude and noise of a BLDC motor in both the time and frequency domains for the X, Y, and Z axes.

Figure 13 presents MATLAB graphical plots depicting vibration amplitude and noise magnitude in the time and frequency domains for the X, Y, and Z axes of a BLDC motor. The vibration amplitude increases at higher RPM, while noise levels decrease in the same RPM range. The signal exhibits periodic behavior with a dominant frequency of approximately 0.4 rad/sample. Higher-frequency components are also present.

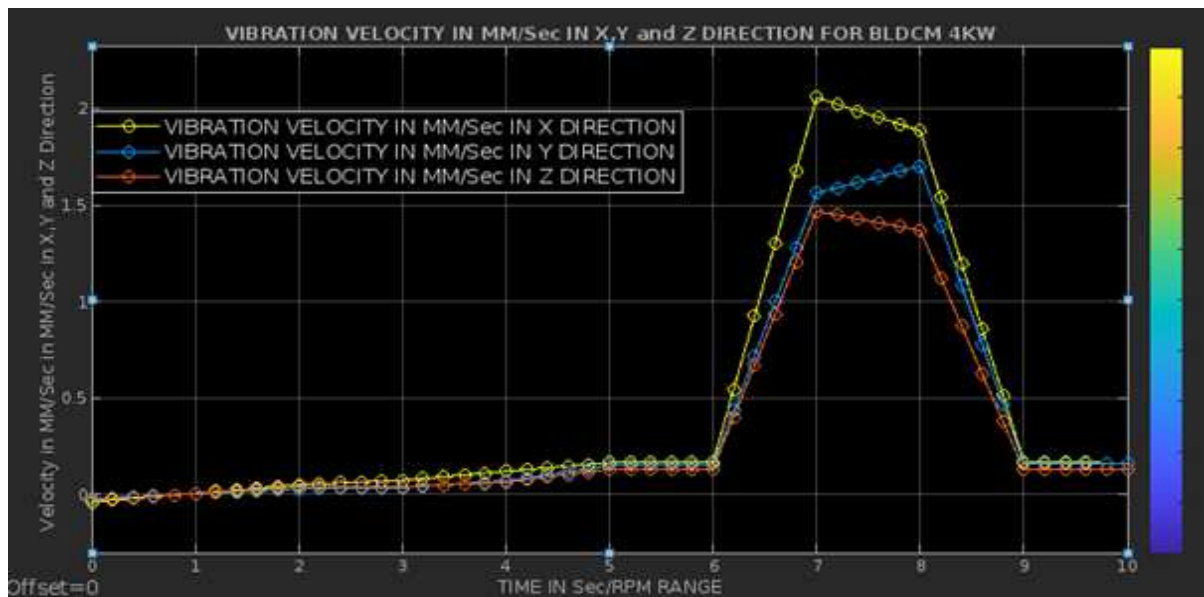


Fig.13. Vibration amplitude and noise of a BLDC motor in both the time and frequency domains for the X, Y, and Z axes.

Figure 14 illustrates the energy and power performance of the BLDC motor across different speed ranges, as observed in a MATLAB/Simulink simulation. The graph shows the trajectory of current, power input, energy, power, torque, mechanical power output, and overall efficiency. Maximum power and energy are reached at around 100 seconds, after which they gradually decrease.

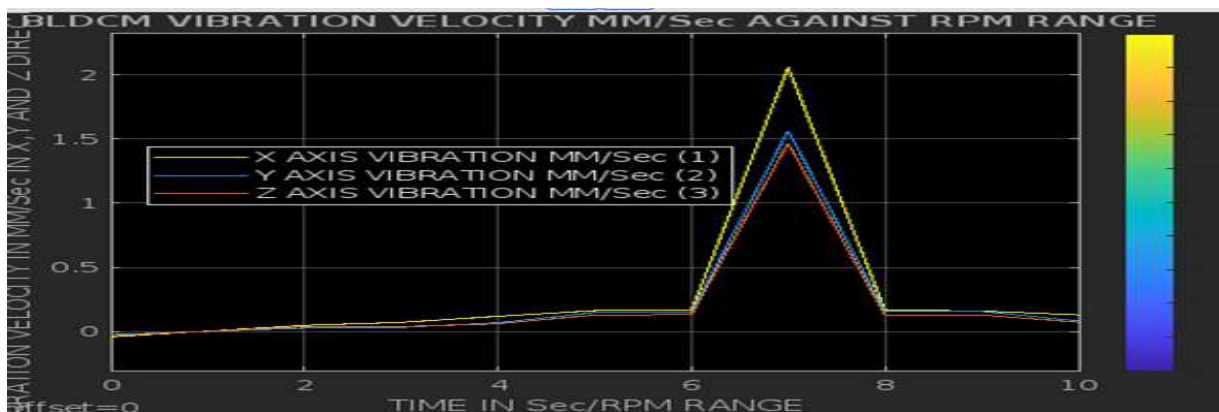


Fig. 14. Energy and Power Performance Curve of a BLDC Motor in a MATLAB/Simulink Simulation

Figure 15. Illustrates the performance of a Brushless Direct Current Motor (BLDCM) in terms of energy and power across the RPM range, spanning from 1280 to 7500 RPM and then from 7500 back to 1280 RPM.

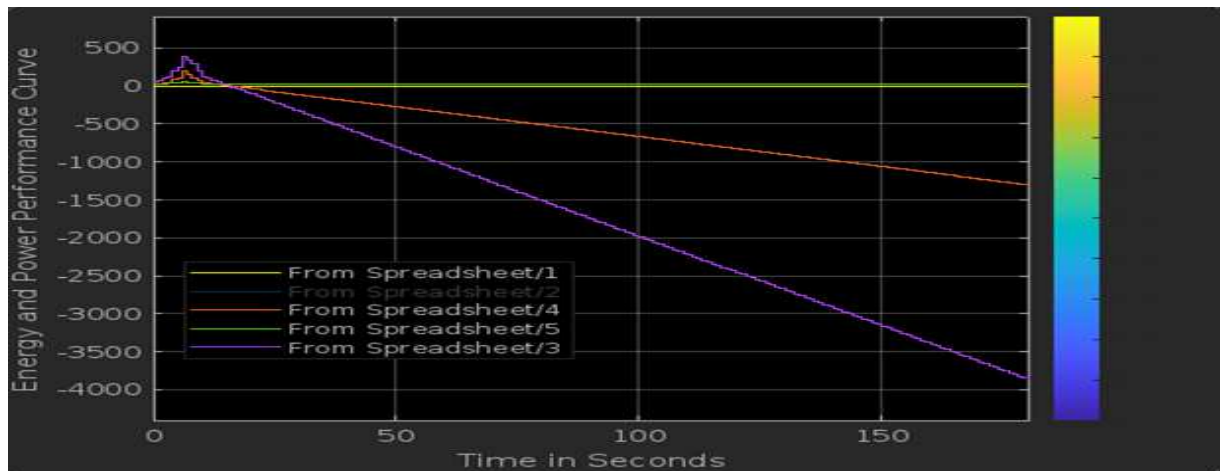


Fig. 15, MATLAB/Simulink simulation results for the BLDC Motor's

Figure 16 displays vibration velocity readings for the BLDC motor at different rotational speeds in the X, Y, and Z directions. Notably, elevated vibration amplitude is observed at 7500 RPM, with the X-axis showing a more rapid increase with RPM. This graph can aid in diagnosing issues with rotating machinery.

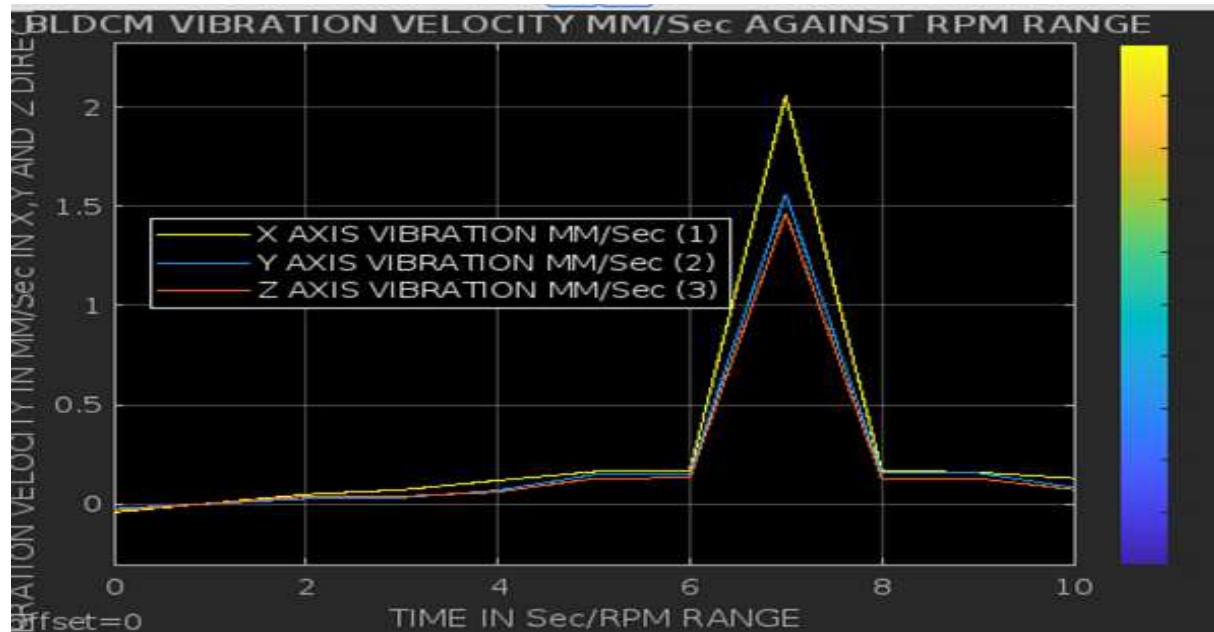


Fig. 16. Trend of Vibration Velocity (in millimeters per second) for a BLDCM (Brushless DC Motor) in the X, Y, and Z directions, sourced from Simulink.

These figures provide valuable insights into the performance and behavior of BLDC motors, particularly regarding vibration, energy, power, and efficiency characteristics. As for the sections on

Implications and Applications: Section 6 explores various implications and applications of the study's findings:

Figure. 17 presents simulation results of the vibration velocity readings (in MM/sec) for a BLDCM (Brushless Direct Current Motor) ranging from 1280 to 7500 RPM, and then back from 7500 to 1280 RPM, in the X, Y, and Z directions. Notably, during this range, it was observed that the vibration amplitude in the X-axis direction exceeded that in the Y and Z axes, particularly between 3800 and 7500 RPM.

The vibration velocity of a 4kW BLDC motor increases in the X-direction as the time or RPM range increases, reaching a maximum at the natural resonance frequency of the motor. The vibration velocity of a 4kW BLDC motor is highest in the X-direction and increases as the time or RPM range increases. The vibration velocity of a 4kW BLDC motor is a good indicator of the health of the motor. Excessive vibration can cause premature wear and tear. The causes of excessive vibration in a motor include an unbalanced rotor, loose bearings, misaligned shaft, worn components, and debris.

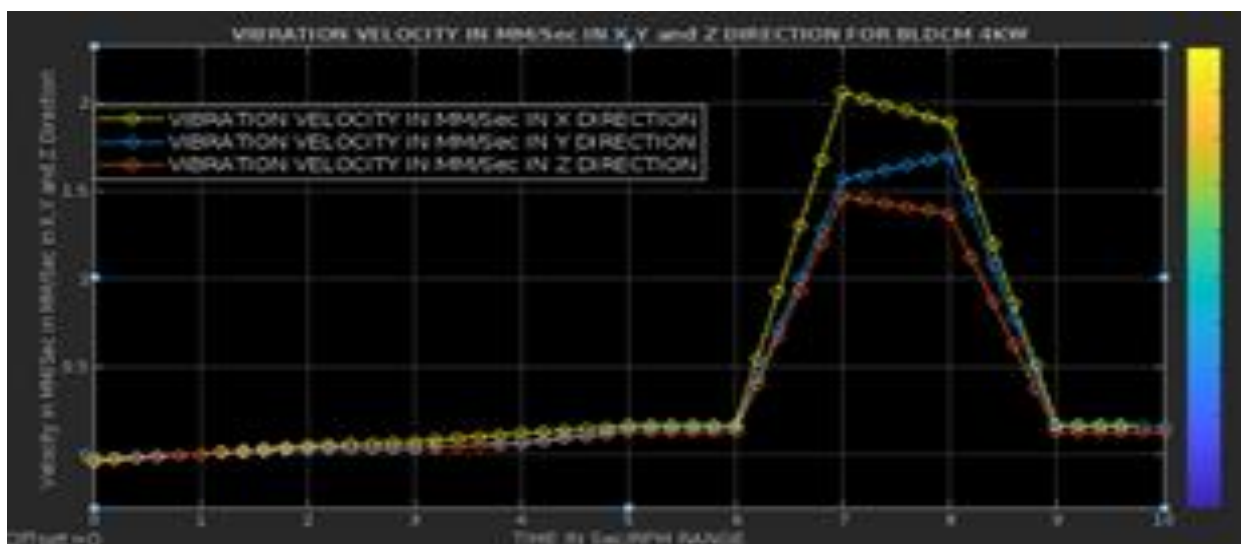


Fig. 17. The vibration velocity (in millimeters per second) of a Brushless DC Motor (BLDCM) in the X, Y, and Z directions, sourced from MATLAB/Simulink.

The MATLAB/Simulink simulation results for the Energy and Power Performance curve of a Brushless DC Motor (BLDCM) spanning the range from 1280 to 7500 RPM and then returning to the RPM range from 7500 to 1280. The simulation illustrates the trajectory of current and power inputs, as well as the resulting Energy, power, torque, Mechanical power output, and overall efficiency across different speed ranges.

Figure 18. illustrates the change in power and performance of a BLDCM motor over time or RPM range. The figure provides a comprehensive overview of the BLDCM motor's power and performance characteristics. This exercise can be used to design and optimize BLDCM motors for specific applications.

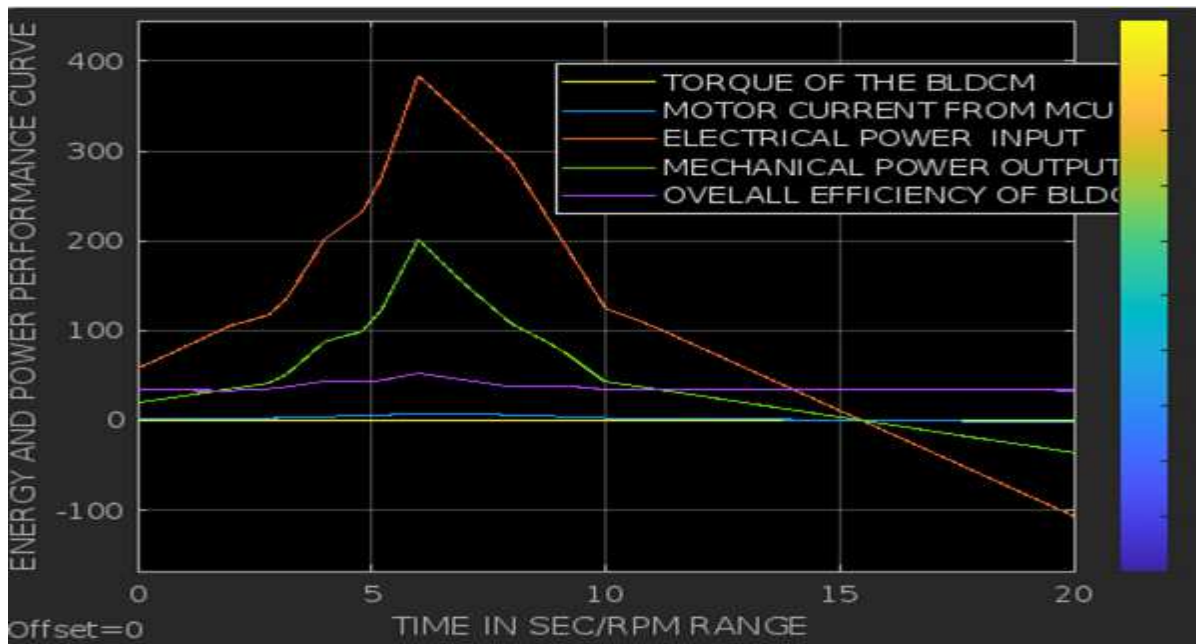


Fig. 18. The performance of a Brushless Direct Current Motor (BLDCM) in terms of energy and power across the speed range, specifically from 1280 RPM to 7500 RPM and from 7500 RPM to 1280 RPM.

5. Results

The research findings are centered on reducing vibrations by identifying noisy motors through waveform analysis of power signals, with a focus on the harmonic content of phase current. Noise induced vibration in the motor can stem from various sources, including electromagnetic factors like cogging torque and torque ripple, mechanical factors such as bearing and rotor vibrations, and the use of field-oriented control (FOC) by the MCU. Analyzing the acoustic trends in BLDC motors for critical analysis poses challenges in hardware and software design. Key constraints in BLDC motor design relate to critical thinking system limitations and parameter measurement.

The study also calculates the overall efficiency of the motor, which is determined by dividing mechanical power by electrical power. The highest overall efficiency coincides with the peak motor efficiency. Additionally, the research explores the relationship between time and RPM range, indicating that the time required for the motor to reach a specific RPM from a standstill increases as RPM values rise, as the motor exerts greater effort to accelerate at higher RPMs. These findings contribute to advancing research in BLDC motor dynamics, the motor's built-in powertrain, CAN Bus communication, and power analysis.

6. Discussion

The research findings highlight the significance of the system's response to a step input in evaluating the system's design. Two critical response types, transient and steady-state responses, were identified. The study employed a simulation model with a BLDC motor speed acting as a step input to determine system parameters and optimize acoustic-vibration effects. The step response analysis revealed a settling time of 20 seconds for the BLDC motor to reach a stable speed, and this duration was found to be necessary for achieving a steady state in response to the input command. The study noted that, initially, the motor exhibited high torque due to low speed, but as acceleration increased, the motor transitioned out of the zero-speed region. The speed eventually reached a steady state, and with decreasing acceleration, favorable noise readings were observed. The simulation further demonstrated that the motor speed closely approached the desired reference speed with minimal error and no overshoot. The correct selection of PI gains was identified as essential for this outcome. The step response of the motor concerning current reflected the desired power and energy profile, with the motor providing power to overcome inertial forces, balancing with power drawn from the battery as it reached a steady state at 4 kW.

The research also considered energy utilization and factors affecting power balance, such as noise, vibration, and heating due to friction. A spectrogram generated from the simulation results illustrated energy dissipation alongside noise factors. This information has practical implications for developing strategies to mitigate acoustic noise in the design phase of BLDC drives, contributing to the optimization of system performance and efficiency.

Research Findings

These figures provide valuable insights into the performance and behavior of BLDC motors, particularly regarding vibration, energy, power, and efficiency characteristics. As for the sections on Implications and Applications: Section 6 explores various implications and applications of the study's findings:

1 Insights for Motor Manufacturers: The study's results offer valuable insights for motor manufacturers, aiding in the optimization of BLDC motors for specific applications and performance requirements.

2. Design Implications for Hybrid Vehicle Chassis: The research outcomes have direct implications for the design and integration of BLDC motors into hybrid vehicle chassis, potentially leading to improved energy efficiency and performance.

3 Implications for Predictive Maintenance: The data and analysis in this study can be applied to predictive maintenance strategies, helping operators detect and address issues in BLDC motors before they lead to critical failures.

4. Future Applications: The study's findings may open up new avenues for future applications of BLDC motors, potentially revolutionizing various industries with improved efficiency and performance

7. Conclusions

In conclusion, this research focused on reducing vibrations in a 2-phase BLDC motor by optimizing the switching time, which was determined by the Hall sensor's position. The study found that this adjustment significantly reduced torque ripples and improved motor performance. It was demonstrated that brushless motors offer advantages over traditional brushed motors, including higher speed, torque capabilities, and lower noise generation. The research emphasized the importance of combining theoretical modeling with experimental data to gain a comprehensive understanding of motor performance. Discrepancies between theoretical and experimental results were attributed to assumptions in calculations and measurement device imperfections. Systematic measurements and optimization efforts were deemed essential to enhance motor performance effectively. This study contributes valuable insights for the design and operation of BLDC motors, with practical implications for various applications.

Future Research Directions

Considering the obtained results and insights from this study, future research in this area could explore advanced control strategies for BLDC motors to mitigate torque variations and improve efficiency. Additionally, further investigations into vibration analysis techniques and their applications in motor diagnostics could lead to enhanced motor performance and durability.

The methodology involved a multidisciplinary approach, combining aspects of mechanical engineering, electrical engineering, and data analysis to comprehensively evaluate the BLDC motor and its performance under real-world conditions.

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Nomenclature

E : (Effective) work potential

E_0 : Exergy

E_{00} : Energy of a system

K : Kelvin temperature scale

S : Entropy

T : Temperature or Celsius temperature scale

W : Effective work

Conflicts of interest

The authors declare that they have no conflicts of interest.

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